

EXTENDS *Naturals*

The *UUID* generation is expected never to be reused. This deliberately models the defensive *ABA* case anyway: a challenge observes *G1*, the state leaves *G1* and later returns to *G1*, and the epoch prevents the delayed challenge from being accepted against the recycled value.

VARIABLES

currentGeneration,
epoch,
observedGeneration,
observedEpoch,
challengeStaged,
challengeUsed,
recycled,
outcome

vars $\triangleq$ $\langle$ *currentGeneration*, *epoch*, *observedGeneration*, *observedEpoch*,
challengeStaged, *challengeUsed*, *recycled*, *outcome* $\rangle$

Init $\triangleq$

$\wedge$ *currentGeneration* = "G1"
 $\wedge$ *epoch* = 1
 $\wedge$ *observedGeneration* = "none"
 $\wedge$ *observedEpoch* = 0
 $\wedge$ *challengeStaged* = FALSE
 $\wedge$ *challengeUsed* = FALSE
 $\wedge$ *recycled* = FALSE
 $\wedge$ *outcome* = "ready"

Observe $\triangleq$

$\wedge \neg$ *challengeStaged*
 $\wedge$ *challengeStaged*' = TRUE
 $\wedge$ *observedGeneration*' = *currentGeneration*
 $\wedge$ *observedEpoch*' = *epoch*
 $\wedge$ UNCHANGED $\langle$ *currentGeneration*, *epoch*, *challengeUsed*, *recycled*,
outcome $\rangle$

RecycleGeneration $\triangleq$

$\wedge$ *challengeStaged*
 $\wedge \neg$ *challengeUsed*
 $\wedge \neg$ *recycled*
 $\wedge$ *recycled*' = TRUE
 $\wedge$ *currentGeneration*' = "G1"
 $\wedge$ *epoch*' = *epoch* + 2
 $\wedge$ UNCHANGED $\langle$ *observedGeneration*, *observedEpoch*, *challengeStaged*,

$\langle challengeUsed, outcome \rangle$

$Confirm \triangleq$
 $\wedge challengeStaged$
 $\wedge \neg challengeUsed$
 $\wedge challengeUsed' = \text{TRUE}$
 $\wedge outcome' = \text{IF } observedGeneration = currentGeneration$
 $\quad \wedge observedEpoch = epoch$
 $\quad \text{THEN "accepted"}$
 $\quad \text{ELSE "rejected"}$
 $\wedge \text{UNCHANGED } \langle currentGeneration, epoch, observedGeneration,$
 $\quad observedEpoch, challengeStaged, recycled \rangle$

$Next \triangleq Observe \vee RecycleGeneration \vee Confirm$

$Spec \triangleq Init \wedge \Box [Next]_{vars}$

$TypeOK \triangleq$
 $\wedge currentGeneration = \text{"G1"}$
 $\wedge epoch \in \{1, 3\}$
 $\wedge observedGeneration \in \{\text{"none"}, \text{"G1"}\}$
 $\wedge observedEpoch \in \{0, 1\}$
 $\wedge challengeStaged \in \text{BOOLEAN}$
 $\wedge challengeUsed \in \text{BOOLEAN}$
 $\wedge recycled \in \text{BOOLEAN}$
 $\wedge outcome \in \{\text{"ready"}, \text{"accepted"}, \text{"rejected"}\}$

$RecycledChallengeIsRejected \triangleq$
 $(recycled \wedge challengeUsed) \Rightarrow outcome = \text{"rejected"}$
